IndianOil	LSTK PACKAGE-2 OFFSITE AND UTILITIES FOR "ETHYLENE GLYCOL" AT PARADIP REFINERY, ODISHA	L&T Hydrocarbon Engineering
TOYO ENGINEERING	Doc. No. : IPEG-6389-8850-PR-002-0002 Title : Document Numbering Procedure	Rev. No. : A2

OWNER	:	**INDIAN OIL CORPORATION LIMITED (IOCL)**
PMC	:	**TOYO ENGINEERING INDIA PRIVATE LIMITED (TEIPL)**
LSTK CONTRACTOR	:	**L&T HYDROCARBON ENGINEERING LIMITED (LTHE)**
UNIT / UNIT NO.	:	**COMMON / 002**
PMC JOB NO. / NAME	:	**6389 / IPEG**
LTHE JOB NO.	:	**IOCL-O&U**

Drawing / Document Category (Submission Purpose)	Drawing / Document Review Status (By OWNER / PMC)		
	Approval Code	Status	Action Required
✓ For Approval	1	Rejected	Revise and resubmit
	2	Commented as noted	Work may proceed subject to acceptable and incorporation of comments
For Review	3	Acceptable with minor comments	Incorporate comments, if any. Resubmit as certified final. Work may proceed.
	4	Approved with No comments – FINAL	Work may proceed.
For Information	5	Information only	Accepted for information only. Work may proceed.

Name	Discipline	Sign	Date

Rev.	Description	Date	Prepared By	Reviewed By	Approved By
			Satyam	Kj	
A2	For Approval	28/02/2019	SY	AJ	JS
A1	For Approval	13/02/2019	SY	AJ	JS

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CHANGE LIST

SL. NO.	REV. NO.	PAGE NO.	CLAUSE / LINE NO.	BRIEF DESCRIPTION / REASON OF REVISION	REMARKS
1	A1	-	-	Issued for Approval	-
2	A2	-	-	Subsequent engineering quick off meeting	-

A Revision line in left side margin denotes revision

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1 SCOPE

IOCL is planning for installation of Ethylene Glycol production facility comprising of an ERU (Ethylene Recovery Unit) of capacity 180KTA, Glycol unit of capacity 357 KTA (comprising of MEG – 332 KTA, DEG-24 KTA and TEG-1 KTA), Intermediate and Product Storage, Dispatch facilities & Utility facilities at Paradip Refinery Complex.

Toyo Engineering India Pvt. Ltd. (TEIPL) has been selected by OWNER as their PMC for this Project.

The proposed Refinery site is located approximately 7 km South West of Paradip Port on the North bank of the River Kansarbatia and straddling the River Santra approximately 2 km from the shore line. The Project site is located close to the coast line. The shore runs in the northeast – southwest direction parallel to the prevalent sea current.

Project has been separated with three LSTKs. The details of LSTKs are as follows

LSTK – Package – 1 – Ethylene Recovery Unit & Ethylene Glycol Unit

LSTK – Package – 2 – Off site Facilities & Utilities

LSTK – Package – 3 –MEG Dispatch Facilities, Jetty and Associated Work

This document is related to LSTK -2 package



2 DEFINITION

OWNER	-	M/s Indian Oil Corporation Limited (IOCL)
PMC	-	TOYO Engineering India Private Limited (TEIPL)
CONTRACTOR	-	L&T Hydrocarbon Engineering Limited (LTHE)
DETAIL ENGG CONTRACTOR	-	L&T-Chiyoda Limited
PM	-	Project Manager
PEM	-	Project Engineering Manager
LE-FD	-	Lead Engineer-Functional Discipline
DC	-	Document Controller
IDC	-	Inter-Disciplinary Checks
EDMS	-	Electronic Document Management System
CRS	-	Comment Resolution Sheet
MDR	-	Master Deliverable Register
VDR	-	Vendor Document Review



3 REFERENCES

Document Numbering Procedure : IEGP-6351-DNP-003 Rev. T1

4 PURPOSE

The procedure outlined in this document specifies the methodology of document numbering for documents / drawings generated w.r.t LSTK-2 package (Offsites and utilities) of Ethylene Glycol Project at IOCL Paradip, refinery by LSTK-2 Contractor including vendor documents.

5 RESPONSIBILITIES

The allocation and control of project document & drawing numbers shall be the responsibility of each Lead Discipline Engineer/Group Lead. Monitoring that this procedure is being correctly implemented shall be the responsibility of the Project Engineering Manager.

6 PROCEDURE

All documents & drawings shall be numbered in accordance with this procedure unless otherwise approved in writing by IOCL. Drawing & document numbers shall not be duplicated under any circumstances. Once the appropriate numbers are assigned and issued, they shall not be reused.

7 NUMBERING SYSTEM:

7.1 Numbering System for documents/drawings submitted to OWNER/PMC:

Numbering System for Project Drawings / Documents/ Requisition / Vendor Documents / Demolition Drawings… to be submitted to OWNER/PMC will be followed as per below document numbering system:

i) The structure for **document/drawing numbering for new facilities** submitted to OWNER/PMC except vendor drawings will be as follows:

Project (1) *	Contract Identifier (2) *	Discipline Identifier (3) *	Document/Drawing Type (4) *	Plant/Unit Number (5) *	Sequential Number (6) *
Alphabetic	Numeric	Numeric	Alphabetic/Numeric	Numeric	Numeric
IPEG	AAAA-	BBBB-	CC-	DDD(D)-	FFFF
Mandatory	Mandatory	Mandatory	Mandatory	Optional (Refer Description)	Mandatory

* For details refer Table 1

Example- New facility Drawings/Documents numbering for Engineering Flow Diagram (Process Engineering) in Storage area is as indicated below:

1	2	3	4	5	6
Project	Contract Identifier	Discipline Identifier	Document/Drawing type	Plant/Unit Number	Sequential Number
IPEG	6389	8110	25	207B	0001

ii) The structure for **vendor documents/drawings numbering for new facilities** submitted for OWNER/PMC will be as follows:

Project (1) *	Contract Identifier (2) *	Vendor Code (7) *	Discipline Identifier (3) *	Document/Drawing Type (4) *	Plant/Unit Number (5) *	Sequential Number (6) *
Alphabetic	Numeric	Alphabetic	Numeric	Alphabetic/Numeric	Numeric	Numeric
IPEG	AAAA-	VVVV(1)-	BBBB-	CC-	DDD(D)-	FFFF
Mandatory	Mandatory	Mandatory	Mandatory	Mandatory	Optional (Refer Description)	Mandatory

* For details refer Table 1

Example - Vendor Drawings/Documents numbering Engineering Flow Diagram (Process Engineering) in Storage area submitted by Flowserve Vendor is as indicated below:

1	2	7	3	4	5	6
Project	Contract Identifier	Vendor Code	Discipline Identifier	Document/Drawing type	Plant/Unit Number	Sequential Number
IPEG	6389	FLWV(1)	8110	25	207B	0001

iii) The structure for **demolition documents/drawings** numbering for OWNER/PMC will be as follows:

Existing Drawing Number	Demolition/Modification (8) *	Project (1) *	Contract Identifier (2) *
Alphabetic/Numeric	Alphabetic	Alphabetic	Numeric
----	GG[GG]-	IPEG	AAAA-
Mandatory	Mandatory	Mandatory	Mandatory

* For details refer Table 1

Example – For numbering of Existing P&ID of Polypropylene plant Offsite lines(Utilities) – Flare Stack Package, where new lines are to be added, demolition drawing number will be provided as below:

	8	1	2
Existing Drawing No	Demolition/Modification	Project	Contract Identifier
PP0S-8110-25-071-0059	DM/RV	IPEG	6389



TABLE-1

(1) Project :	All document/drawing number shall start with the project Identifier **"IPEG"**
(2) Contract Identifier :	**AAAA - Contract Identifier** - Unique contract number allocated for this project is **6389.**
(3) Discipline Identifier :	**BBBB - Discipline Identifier** This is a four (4) - digit number, which shall be the "owning" Group / Discipline code, refer **Attachment 1**.
(4) Document/Drawing Type :	**CC - Document / Drawing Type** This is a two (2) digit code. Document is denoted by a 2 digit letter code Drawing is denoted by a 2 digit number code For a Document / Drawing Type refer to **Attachment 2 (A) & 2 (B).**
(5) Plant / Unit Number :	**DDD(D)- Plant/Unit Number** This is a four (4) digit code equal to the Unit number, to which the document / drawing is applicable. For Plant/Unit Number refer to **Attachment 3** **Note - 1) Additional fourth letter will be used for all documents/drawings wherein four digit code is indicated for plant / unit / buildings as per ITB (E.g. Cooling Tower substation building – 316N** **Note – 2) Equipment Tags are to be prefixed with the respective Unit numbers. Thus, Unit numbering is applicable for tag of equipment. '207' as common unit number for all O&U equipment under sections 207A to 207D is to be followed.** **Note -3) P&IDs/ PFD number shall be based on unit numbers. Common number '207' to be followed for unit 207A to 207D.**
(6) Sequential Number	**VVVV - Sequential Number** This is a four (4)-digit number, which shall be from the series 0001 to 9999.

2

(7) Vendor Code :	**EEEE(1) – Vendor Code** Discipline to identify any four digit alphabetical short form of the vendor. For multiple orders on the same vendor, additional one numeric digit to be added. Eg. VVVV1, VVVV2 Suppose LMF (vendor short name) receives 3 orders for items such as reactors, columns and vessel. Then L&T-MFF to follow; LTMF (1) - for reactor LTMF (2) - for columns LTMF (3) - for vessels If vendor receives one order only, Vendor name to be followed by (1) i.e. LMF (1)
(8) Demolition :	**GG[GG] – Demolition/Modification** Document/drawing affected by demolition/Modification works resulting from modification activity will be identified with this section. If separate sheets for documents/drawings are prepared for Demolition and Revamp / modification works, in that case “GG” will be indicated by **DM** & **RV** respectively If Demolition and revamp related mark up in provided only in single sheet / document, “GG[GG]” will be indicated by **DMRV**. Revision philosophy for demolition/revamp documents will be in line with new documents
Sheet Number :	The use of sheet numbers shall be kept to a minimum. Where the use of sheet numbers cannot be avoided they shall take the form of a three (3) digit number and shall be depicted as Sht. XXX of YYY and be shown on the Document / Drawing local to the Document / Drawing No. Where sheet numbers are used on Drawings each sheet shall show the full Drawing number as well as the sheet number. Where appropriate, the first sheet number of a multi - sheet drawing shall be 001 and shall always be an index. This index shall list all sheet numbers and the revision history for each sheet.

2

Documents / Drawings Revision Codes :	For documents/drawings revision codes refer **Attachment 4.**
Documents/Drawings Revision Indicators	Changes to documents shall be indicated by highlighting **(revision mark, cloud mark,traingle,underline text)** the relevant text and the following statement shall be made on the front sheet: **'Changes to this document are highlighted as per this statement'** For details on control of documents refer Project Execution Procedure(Document No :IPEG-6389-8820-PR-000-001)

8 ATTACHMENTS

ATTACHMENT 1 - DISCIPLINE CODES

Sr. No.	Code No.	Title
1	8110	Process Engineering
2	8120	Process Control
3	8130	Commissioning and Operations
4	8140	Fire Engineering
5	8150	Environmental Engineering
6	8210	Pipelines

Sr. No.	Code No.	Title
7	8230	Piping Engineering
8	8310	Civil Engineering
9	8320	Structural Engineering
10	8330	Architecture
11	8340	Building Services
12	8380	Sub Structures and Sub-sea Engineering
13	8390	Weight Control
14	8410	Rotating Equipment
15	8420	Mechanical Handling
16 2	8430	Static Equipment
17	8440	Technology
18	8470	Package Equipment
19	8530	Electrical Engineering
20	8531	Electrical Engineering Documents
21	8532	Electrical Engineering Drawings
22	8540	Telecommunications
23	8541	Telecommunications Documents
24	8542	Telecommunications Drawings
25	8551	Instrumentation Documents
26	8552	Instrumentation Drawings
27	8560	Loss Prevention
28	8610	Materials Management
29	8620	Purchasing Engineers and Buyers
30	8630	External Expediting
31	8640	Inspection and Quality Surveillance
32	8650	Transportation

Sr. No.	Code No.	Title
33	8710	Home Office Construction
34	8720	Construction Contracts
35	8730	Quantity Surveying
36	8820	Project Management and Engineering
37	8830	Quality Assurance
38	8840	Project Control
39	8850	Document Control
40	8860	Estimating
41	8870	Project IT
42	8890	Others (Accounting)
43	8920	Commercial

ATTACHMENT 2 (A) - FOR DESIGN DOCUMENT TYPE CODES

Sr. No.	Code No.	Title
1	AE	Authorized Engineers List
2	BM	Bill of Material
3	BQ	Bills of Quantity / Material Take-off
4	CA	Calculations
5	CE	Contract Engineering Data Sheet
6	CO	COSHH Assessments
7	CP CS	Contract Package. Construction Strategies
8	DI	Discipline Design Instructions / Design Basis / Design Criteria
9	DK	PC Discs (including back-ups)
10	DM	Distribution Matrix
11	DS	Data Sheets

Sr. No.	Code No.	Title
12	EL	Equipment Lists
13	FL	Flysheet
14	FM	Forms
15	FQ	Field Queries (answered at site)
16	FS	Functional Specifications – Major Control Systems
17	GN	General Notes for Requisitions
18	HD	Handover Dossiers (site only)
19	HR	HAZOP Reports
20	HS	Hazardous Material Schedule
21	IN	Document/Drawing Indexes (not for lists e.g. see Line Lists in attachment 2)
22	MS	Method Statements
23	OM	Operating Manual
24	PD	Process Documents
25	PH	Philosophy Documents
26	PI	Project Instructions
27	PL	Planning Documents
28	PO	Purchase Orders
29	PR	Procedures
30	PS	Process Specifications
31	QA	QA Documentation
32	QR	Quotation Requests
33	RA	Risk Assessments
34	RP	Reports
35	RQ	Material Requisitions
36	SA	Safety Documentation
37	SC	Contract Requisitions
38	SH	Schedules (use 'PL' for Level 1 to 4 schedules)

Sr. No.	Code No.	Title
39	SI	Special Piping Items
40	SL	Supplier list
41	SP	Project Specific Engineering Specifications & Standards. (Use TN for scope of work and similar narratives)
42	SQ	Site Queries (sent to Home Office for resolution)
43	ST	Studies
44	TN	Technical Notes
45	TP	Test Plan
46	TQ	Technical Queries
47	VS	Instrument Vessel Sketches
48	YX	Licensor Information
49	YY	IOCL Documents
50	YZ	IOCL Specifications/Standards
51	TB	Technical Bid Evaluations
52	CH	Charts / Graphs
53	SR	Stress Analysis Report
54	OD	Other Documents
55	LD	Foundation load data

2

2

ATTACHMENT 2 (B) – FOR DESIGN DRAWINGS TYPE CODES

Sr. No.	Code No.	Title
1	1	Site Plans, Site Layouts, Plot Plans & Area Plans
2	2	Key Plans
3	3	Permitting and Planning Approval Drawings
4	11	Static Equipment - Towers & Trays
5	12	Static Equipment - Reactors & Regenerators
6	13	Static Equipment - Drums
7	14	Static Equipment - Tankage
8	15	S&T Heat Exchanger Drawings
9	16	A/C Heat Exchanger Drawings
10	19	Static Equipment - Miscellaneous
11	20	Process Flow Diagrams
12	21	Material Balance Diagrams

Sr. No.	Code No.	Title
13	22	Pressure & temperature Diagrams
14	23	Material Selection Diagrams
15	24	Cause & Effect Diagrams
16	25	Engineering Flow Diagrams (EFDs or P&IDs)
17	26	Line Lists
18	27	Process Control Drawings
19	28	Manufacturing Block Diagrams (for Pharmaceutical use only)
20	29	Fire Fighting Drawings
21	30	Planning Drawings (Networks Etc)
22	39	Civil / Structural Standard Drawings (Anchor Bolts, Standard Steelwork Notes Etc)
23	40	Civil Survey Drawings
24	41	Site Preparation Drawings (Includes Earthworks, Roads and Fencing layouts/details and Temporary Facilities)
25	42	Piling Drawings
26	43	Civil Substructures & Foundations (Includes GA's, RC Details, Foundation Loading Diagrams)
27	44	Concrete Structures (Includes GA's / RC Details)
28	45	Underground Services (Includes Paving & Underground Schematics, U/G Services Layouts & Details, Floor Drainage Layouts & Details)
29	46	Steelwork Structures (Includes Steelwork GA's/Details, Fireproofing GA's/Details)
30	47	Architectural (Includes Plans, Sections, Elevations, Internal & External Details)
31	48	Building Services & HVAC (Includes HVAC EFD's Control Schematics, GA's & Details, Room Pressure Diagrams)
32	49	Building Structures
33	50	Civil layouts
34	51	Piping General Arrangements
35	52	Piping Isometrics (Includes Isometric Index)
36	53	Piping Tie-ins (Includes Tie-in Schedules & Details)

Sr. No.	Code No.	Title
37	54	Miscellaneous Piping Details and schedules
38	55	Steam Tracing Drawings (Includes GA's & Details)
39	56	Standard Drawings (other than Civil / Structural)
40	57	General Arrangement Drawings (other than Civil / Structural & Piping)
41	58	Detail Drawings
42	59	Layout drawings
43	60	Block Diagram
44	61	Loop Diagram
45	62	Logic Diagram
46	63	Single Line Diagram
47	64	Schematic Diagram
48	65	Wiring Diagram
49	66	Other Drawings / Diagrams
50	67	Fabrication drawing
51	68	Nozzle details drawings
52	69	Hook ups
53	70	Graphics
54	71	System configuration diagram

2

2

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ATTACHMENT 3 - PLANT / UNIT NUMBERING INDEX

UNIT NAME	UNIT NUMBER
COMMON	002
COOLING TOWER	65
ETYLENE STORAGE SPHERES & BOG PACKAG	207A
MEG / TEG / DEG STORAGE AREA	207 B
TANKER LOADING & UNLOADING AREA	207 C
TEG BARELLING FACILITY	207D
MEG OFFSITES (OSBL DRAWINGS …)	215
PP FLARE	WILL BE INFORMED LATER
ETP	WILL BE INFORMED LATER

Unit numbering for facilities

Building	Numbers
Cooling Tower Substation	316N
BOOT-3 Substation	317N
SRR (CT)	824
SRR (BOOT)	827
S&D Building	442
COOLING TOWER –CHEMICAL HOUSE	444

2

TEG BARRELLING FACILITY	207D
SAMPLE RETENTION ROOM	443
FIRE WATER BOOSTER PUMP HOUSE	127C
EXTNESION OF PP UNIT LAB BUILDING FOR MEG	WILL BE INFORMED LATER
CAR PARKING	WILL BE INFORMED LATER
LADIES/GENTS TOILET	WILL BE INFORMED LATER

2

ATTACHMENT 4 – REVISION CODES FOR DOCUMENTS/DRAWINGS

Revision Codes for documents/drawings submitted to OWNER/PMC shall be as follows:

Approval Category Documents	**Review Category Documents**	**Information Category Documents**
A1, A2... Issued for approval	O1, O2... Issued for Review	I1, I2... Issued for Information
H1, H2... Issued for HAZOP	H1, H2... Issued for HAZOP	C1, C2... Issued for Construction
D1, D2... Issued for Design	D1, D2... Issued for Design	Z1, Z2... As-Built
C1, C2... Issued for Construction	C1, C2... Issued for Construction	
Z1, Z2... As-Built	Z1, Z2... As-Built	